Next, we split the combined data into train, validation, and test sets using a 50:25:25 percent split for the adversarial and 70:15:15 for the non-adversarial subsets. This boosts the representation of adversarial samples in the overall training set. We train a classifier using the Light Gradient-Boosting Machine (LightGBM) algorithm. We then run predictions on all validation samples and tune the thresholds that map the results to positive (adversarial) or negative (non-adversarial) classes to maximize the F_2 score. The value of β in F_β can be tailored to the particular policy goals of a researcher or practitioner.

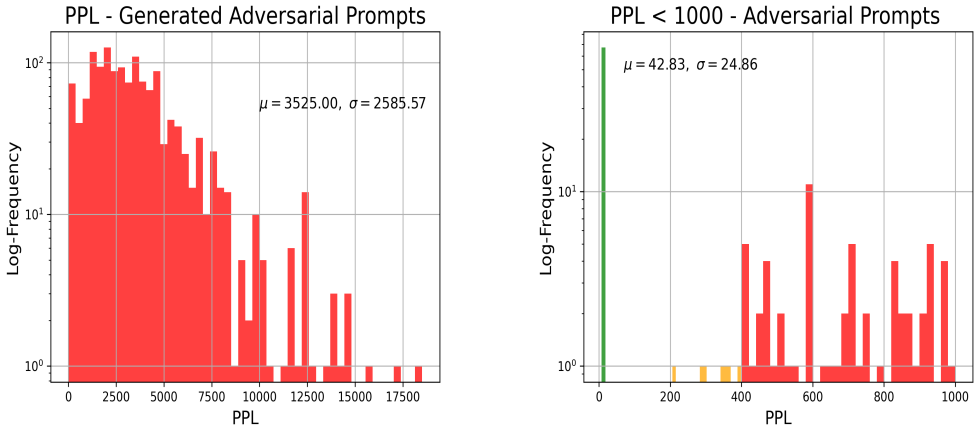
Finally, we run predictions on the test dataset and display a scatter plot that shows the True Positives, False Positives, and False Negatives. Our analysis looks into these results in more detail.

4 DATA

4.1 ADVERSERIAL PROMPT PREPARATION

In this research, we utilized two distinct datasets, each comprising adversarial prompts. The first dataset consists of 1407 machine-generated adversarial prompts, which were generated using the methodology outlined in Zou et al. (2023). Specifically, we employed the Vicuna-7b-1.5 model and executed the "individual" GCG (Greedy Coordinate Gradient) method as described in the aforementioned paper. Our prompts were derived from a randomly selected set of 67 objectives listed under "Harmful Behaviors" in the provided code. For each of these objectives, the GCG model generated 20 distinct attack suffix predictions. Each prompt was initialized with the default *starter* suffix, consisting of 20 spaced exclamation marks, which we found to be benign in our experimental tests. Detailed information about this dataset, including perplexity distributions and sequence lengths, can be found in Appendix A.1.

The generation process for these prompts necessitated 50 hours of computational resources and was conducted on an A100 GPU equipped with 40GiB of memory.



(a) All adversarial prompt counts by perplexity

(b) Low perplexity adversarial ranges

Figure 1: Adversarial prompts based on Zou et al. (2023) exhibit high perplexity. Figure b on the right, shows the lower range of values, color-coded in three distinct clustera. The green, contains all repeat-! suffix attacks with a perplexity below 18. In red, the larger cluster above 400, and in yellow, the few prompts in between.

4.2 ADVERSARIAL PROMPT CLUSTERS

As depicted in Figure 1, the perplexity values of our generated adversarial attacks are notably high. The green cluster comprises prompts with a starter-default suffix, characterized by 20 spaced exclamation marks. Our understanding is that these exclamation mark examples are carryover dummy values used in the initialization of the attack algorithm that can be ignored. In contrast, the clusters on the right contain candidate attack prompts for further evaluation. The perplexity values would

likely be lower if the attack generation process were reconfigured like in Jain et al. (2023) (see related work). It could also vary with additional training time.

4.3 Human-Designed Adversarial Prompts

The second dataset consists of 79 human-designed adversarial prompts tailored for GPT-4 jailbreaking, as outlined in Jaramillo (2023). Further details about this dataset can be found in Appendix A.2. Notably, unlike the machine-generated prompts, these human-designed prompts exhibit low perplexity scores similar to regular text, which highlights the diversity of adversarial prompt characteristics.

4.4 Non-Adversarial Prompts

To provide a comprehensive context for our analysis, we also included benign non-adversarial prompts for comparison purposes. These non-adversarial prompts were sourced from various datasets and conversational contexts, as outlined below:

- 6994 prompts from humans with GPT-4. (see Appendix B.5).
- 998 prompts from the DocRED dataset (see Appendix B.1).
- 3270 prompts from the SuperGLUE (boolq) dataset (see Appendix B.2).
- 11873 prompts from the SQuAD-v2 dataset (see Appendix B.3).
- 24926 prompts with instructions from the Platypus dataset, which were used to train the Platypus models (see Appendix B.4).
- 116862 prompts derived from the *"Tapir"* dataset by concatenating instructions and input (see Appendix B.6).
- 10000 instructional code search prompts extracted from the instructional_code-search-net-python dataset (see Appendix B.7).

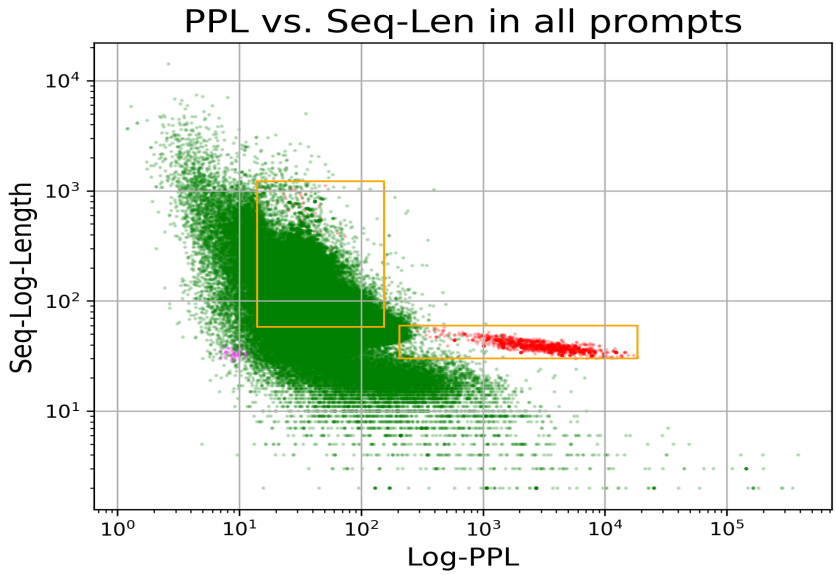
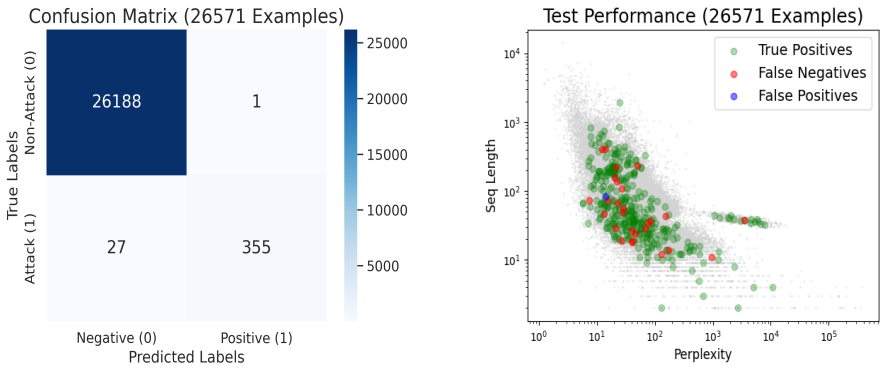


Figure 2: All prompts are shown. The red are adversarial attacks, both machine-generated and human-crafted. The green are regular prompts, while the magenta are machine-generated with repeat exclamation marks as their suffixes.



(a) Predictions on Test Data

(b) TP, TN and FN Predictions

Figure 3: Results from running the classifier on test data. The classifier is highly effective at separating real prompts from adversarial suffix attacks. The human-crafted prompts are responsible for the overwhelming majority of false negatives and false positives.

5 ANALYSIS

The separation between non-adversarial and adversarial prompts in Figure 2 underscores the effectiveness of perplexity as an initial discriminatory metric. This graph also highlights a potential source of false positives in the use of perplexity filtering in plain perplexity filtering, given that some regular prompts overlap significantly on the perplexity axis with adversarial prompts (and even exceed them by an order of magnitude).

Figure 2 shows two orange rectangular regions. The lower right is the region that contains machine-generated prompts based on Zou et al. (2023). The opportunity to separate these prompts from regular prompts is visually evident. The magenta points on the left (looking up from 10^1) represent the prompts with repeat exclamation marks as their suffixes. They have not been shown to penetrate LLMs alignment defenses (and could be easily memorized by a defense) – so they are ignored.

The larger upper-left rectangular region in Figure 2 contains all manually crafted jailbreak prompts by humans, as described in Appendix A.2. Notably, this jailbreaking dataset contains very few data points. Visual inspection shows that these points are disguised among ordinary green non-adversarial prompts in the same region.

Non-adversarial prompts using proper English generally exhibit low perplexity values relative to adversarial. Inspecting the individual regular prompt data sets in the Appendix, we can see the cases where high perplexity scores can occur. Single words or short phrases within a conversational context show an elevation of perplexity scores. In particular, this occurred when non-English tokens, mathematical expressions, programming code, misspelled English, or irregular symbols were involved.

Running the GBM classifier over the test data set yields an F_2 score of 95.6% on the validation set, and 94.2% on the test set is shown in Table 2. F_2 puts more weight on recall than precision. The confusion matrix is shown in Figure 3a, confirming the apparent ability of the classifier to distinguish adversarial suffix attacks from ordinary prompts. A closer look at the failed examples in Table 1 reveals that all false negatives originate from human-crafted jailbreaks. After excluding the human-crafted observations from the results, the F_2 score climbs to 99.1% from 94.2%.

Finally, we contrast the two-feature classifier with a simple approach of defining a perplexity threshold above which we consider the prompt to be an attack. The optimal threshold that maximizes the F_2 score for this simple method is 997, which was the same to several decimal points as 1000. Table 2 contains our test on two threshold values, 400 and 1000, along with the score achieved with the two-feature classifier.